Exp no:01
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Study of Various Network commands used in Linux and windows

Basic Networking commands:

arp -a : ARP is a short form of address resolution protocol. It will show the IP address and MAC address of your router.

arp -a

Output:

-gateway (10.0.2.2) at 52:54:00:12:35:00

hostname. This is the simplest of all TCP/IP commands. It simply displays the name of your computer

Output:

Localhost. localhost. localdomain

ipconfig /all: This command displays detailed configuration information about your TCP/IP connection including Router, Gateway, DNS, DHP, and type of Ethernet adapta in your system

ipconfig

Output:

Windows IP configuration,

Ethernet adapter. Ethernet 2:

Connection-specificatic DNS Suffi

Link-Load IP's address: fe80:60

73f8:684d:a4be1:3

IPV4 Address.... 192.168.56.1

Nbtstat -a : This command helps solve problems with NetBIOS name resolution. (Nbt stands for NetBIOS over TCP/IP)

Output:

Displays protocol statistics and current TCP/IP connections using NBT

BTSTAT [[-a RemoteName] [-A IP address] [-c] [-n] [-r] [-R] [-RR] [-s] [-S] [interval]

netstat: (network statistics) netstat displays a variety of statistics about a computer active TCP/IP connecting. It is a command line tool for monitoring network connectors both incoming and outgoing, as well as viewing statistics etc.

Output:

Active connection

Proto Local address Foreign Address State

TCP 10.55.47.189.50357 4.213.25.310. https.

TCP 10.65.47:189:50358 4.213.25.240 https established

nslookup : It is a tool used to perform DNS lookup in linux. It is used to display DNS details, such as the IP address of a particular computer the MX record for a domain

Output

Default Server : unknown
Address : 10.6.47.646

Pathping : pathping is unique to windows and is basically a combination of the ping and tracert commands

Output :

pathping [-g host -rost] [-h maximum -hops] [-i address] [-n] [-p period] [-q num -queries] [-w timeout] [-4] [-6] target-name

Ping : [Packet Internet grouper)

command is the host way to test connecting between two nodes

Output : usage Ping [-t] [-a] [-n count] [-l size] [-f] [-i ttl] [-v tos] [-r count] [-s count] [-j host-list]

Route : Route command is used to show manipulate the IP routing table

Output:
Kernel IP routing table
Destination Gateway Genmask Flags
Metric Ref 10.0.2.0 0.0.0.0 255.255.255.
V 100 0

Some important linux networking commands:

ip: The ip command can show address information, manipulates, routing, plus displays network various devices, interfaces and tunnels

i) ip address show

Output:
1: lo: <LOOPBACK, UP, LOWER_UP> mtu.
65536 qdisc noqueue state UNKNOWN
group default qlen 1000

(ii) To assign an IP to an interface for example: enp503

ifconfig: The ifconfig command was/is a staple in many sysadmin's tool, belt for configuration and troubleshooting between network:

Output:

flags = 4163 <UP, LOOPBACK, RUNNING, MULTICASIS mtu 1500 unit 10.0.215 network 2555.255.2255.0..

mtr: MTR is a program with a command line interface that server as a network diagnolic and troubleshooting tool. This command combines the functionality of the ping and troubleshooting tools

Output: My traceroute [v0.94]

Host

~~1. localhost~~ Packets Pings

localhost Loss1: Snt Last Avg Best Wrst Stdev

0.0% 122 0.1 0.2 0.1 1.9 0.2

a) mtr google.com

Output: host

1. Gateway

2. (waiting for reply)

b). mtr : g google.com

Output:

Usage: mtr [options] hostname

-F, --- filename FILE read host from a file

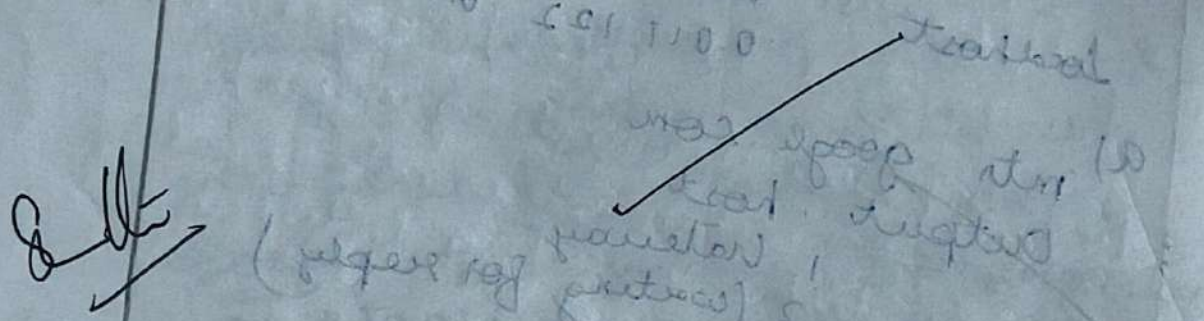
tcpdump: The tcpdump command is designed for capturing and displaying packets

Output : tcpdump : enp0s3 : you dont have permission to capture on that device

Ping : Ping is a tool that verifies IP-level connectivity to another TCP/IP computer by sending internet control message protocol (ICMP) echo request messages.

a). Ping google.com

Output : Ping google.com (142.250.182.78) 56(84) bytes of data 64 bytes from lcmaa-an-in-f14.1e100.net (142.250.182.78) icmp_seq=1 ttl=112 time=938 ms



Result

Thus the various linux and windows command is executed and observed